

A03 Neural Network Zoo - Reflection Journal

Nichelle:

I enjoyed the Neural Network Zoo assignment because the idea of pairing neural network architectures with animals characteristics made the technical concepts more memorable and easier to visualize. I found our team's GNN and wolf-spider metaphor especially interesting because it connected specific characteristics of a Graph Neural Network to the way an organism senses and responds to its environment. In the comparison developed by our team, nodes can be viewed as individual sensory or processing points, edges as the pathways that connect information, and message passing as the process through which information from surrounding connections influences each node.

What helped me understand GNNs more clearly was the idea that a node does not operate in isolation. Its representation is influenced by the nodes connected to it, just as an animal responds not only to its own internal state but also to signals from its environment. That made the concept of message passing much more intuitive to me. The "spider" part of the metaphor emphasizes connectivity and network structure, while the "wolf" part emphasizes active interaction and responsiveness. Together, the metaphor illustrates that a GNN learns from both the characteristics of individual nodes and the relationships connecting them.

I also appreciated that this assignment showed how analogy can be used as a learning tool. A neural network architecture can initially seem abstract when it is explained only in terms of nodes, edges, layers, and mathematical operations. Connecting those ideas to observable characteristics in the natural world gave me another way to organize the concepts mentally. Rather than memorizing a definition of a GNN, I found myself thinking more about what makes it fundamentally different from neural networks designed for images or sequential data: the relationships between entities are part of the information being learned.

Cesar:

Through the "Neural Network Zoo" project, I gained a better understanding of how deep learning architectures specialize in specific data types, much like species adapting to distinct environments.

Analyzing graph neural networks (GNNs) through this metaphor highlighted how network nodes transmit messages across relational edges, mirroring the social dynamics and collective communication of wolves.

Finally, I realized that there is no single "optimal" neural network; simple multilayer perceptions (MLPs), convolutional layers, and graph-based models solve different problems depending on the nature of the input data whether structured, spatial, or relational graph-base.

Mary Ann:

Comparing the wolf, spider, and wolf spider showed how metaphors can highlight different parts of neural network architectures.

The wolf spider provides the best balance of:

- Relational structure
- Distributed sensing
- Conceptual clarity
- Creative alignment with the Neural Network Zoo

The exercise made GNN architecture easier to understand — and easier to teach to others.

Takeaway: GNNs are powerful because they learn from both the features of individual nodes and the relationships connecting them.

Luiz:

I learned that a GNN uses a graph to represent information. The main parts of a graph are nodes and edges. Nodes represent objects, and edges represent the relationships or connections between the objects.

For our wolf example, each wolf can represent a node, and the relationships between the wolves can represent the edges. Another important part I researched was message passing. This is how the nodes in a GNN share information with their neighbors. Each node receives information from the nodes connected to it and uses that information to update its own information. This process can happen through multiple GNN layers.

I also learned that the number of layers can affect how much information a node can receive from the rest of the graph. With one layer, a node can learn from its direct neighbors. With more layers, information can travel farther through the graph.

This research helped me understand GNNs better. Before working on this assignment, I did not know much about how graph-based neural networks worked. Now I understand that GNNs are focused not only on individual objects, but also on the relationships between those objects